Genetic Factors Influencing Asthma Exacerbation Frequency: A Systematic Review of Candidate Gene Polymorphisms

Abstract

Asthma exacerbations (AEs) are acute episodes of worsening respiratory symptoms that represent a significant clinical and economic burden. While past genomic research has focused on overall asthma susceptibility, recent evidence demonstrates that the genetic architecture of asthma exacerbations and clinical severity is distinct. This systematic review evaluates candidate gene polymorphisms associated with the frequency and severity of asthma exacerbations, including specific SNPs in *ADRB2, IL4, IL4RA, TNFA, NR3C1, PPARG, TGFB1, IL10, CHI3L1, GSDMB, CTNNA3, ADIPOQ*, and *SEMA3D*. Through a systematic search and narrative synthesis of literature published between 2015 and 2025, we investigate the clinical evidence and biological mechanisms underlying these associations. We find robust, replicated clinical evidence for several risk and protective loci governing type 2 inflammation, airway remodeling, pharmacogenetic responses, and epithelial barrier dysfunction, supplemented by functional expression quantitative trait loci (eQTL) data. Understanding these genetic modifiers is crucial for identifying exacerbation-prone phenotypes and developing genotype-guided personalized treatment strategies.

1 Introduction

1.1 Burden of Asthma Exacerbations

Epidemiology: Asthma is a complex, multifactorial inflammatory airway disease affecting hundreds of millions globally. A substantial proportion of morbidity is driven by asthma exacerbations (AEs), which are acute or sub-acute episodes of progressively worsening shortness of breath, cough, wheezing, and chest tightness [1, 2]. **Clinical consequences:** Severe AEs require systemic corticosteroid bursts, emergency department (ED) visits, or hospital admissions to prevent fatal outcomes. Recurrent exacerbations accelerate lung function decline and increase mortality risk [3]. **Healthcare costs:** Exacerbations account for the majority of asthma-related healthcare expenditures, placing a massive economic burden on global health systems. Preventing these episodes is the primary goal of current clinical guidelines [4].

1.2 Biological Mechanisms Underlying Asthma Exacerbations

Airway inflammation: Exacerbations are fundamentally driven by acute amplifications of underlying airway inflammation, often involving both Type 2 (eosinophilic) and non-Type 2 (neutrophilic) pathways [5]. **Viral infections:** Viral respiratory pathogens, notably human rhinovirus C (HRV-C), are the most common triggers, responsible for up to 80% of pediatric exacerbations [6, 7]. **Environmental triggers:** Aeroallergens (e.g., house dust mite), air pollution (PM2.5, NO2), and tobacco smoke exposure interact with host biology to precipitate acute attacks [8, 9]. **Airway remodeling:** Chronic structural changes, including subepithelial fibrosis, smooth muscle hypertrophy, and angiogenesis, increase airway wall stiffness, rendering the airways hyperresponsive to triggers and prone to severe bronchoconstriction [10, 11].

1.3 Genetic Contributions to Asthma Outcomes

Heritability of asthma: Twin and family-based studies reveal high heritability for asthma, particularly childhood-onset disease (estimated at 80-90%) compared to adult-onset asthma [12,13]. **Genetics of asthma severity:** Accumulating evidence demonstrates that the genetic architecture governing asthma exacerbation susceptibility and clinical severity is distinct from that of general disease risk [14,15]. **Genetics of treatment response:** Pharmacogenomic variants modify therapeutic efficacy, contributing to inter-individual variability in response to β2-agonists, inhaled corticosteroids (ICS), and leukotriene modifiers [16,17].

1.4 Candidate Genes Potentially Involved in Asthma Exacerbations

Brief rationale for each candidate gene:

- *ADRB2*: Encodes the β2-adrenergic receptor; polymorphisms affect bronchodilator response and tachyphylaxis [18,19].
- *IL4* & *IL4RA*: Key components of Type 2 inflammation; variants upregulate IgE synthesis and eosinophilic pathways [20,21].
- *TNFA*: Pro-inflammatory cytokine driving neutrophilic inflammation and airway hyperresponsiveness [22,23].
- *NR3C1*: Encodes the glucocorticoid receptor; modulates sensitivity to steroid therapies [24,25].
- *PPARG*: Regulates lipid metabolism and anti-inflammatory signaling; rare variants offer protection [4,26].
- *TGFB1*: Master regulator of airway remodeling and subepithelial fibrosis [27,28].
- *IL10*: Crucial anti-inflammatory cytokine; promoter variants alter regulatory T cell function [4,29].
- *CHI3L1*: Encodes YKL-40, linked to airway remodeling, inflammation, and severity [14, 30].
- *GSDMB*: 17q12-21 locus gene linked to epithelial integrity and viral-induced exacerbations in early childhood [31,32].
- *CTNNA3*: Regulates epithelial adherens junctions; variants increase exacerbation rates [6].
- *ADIPOQ*: Modulates systemic metainflammation, linking obesity to asthma severity [33, 34].
- *SEMA3D*: Regulates angiogenesis, structural remodeling, and immune cell recruitment [6,35].

1.5 Rationale for the Review

Gap in the literature: While genome-wide association studies (GWAS) and systematic reviews have extensively characterized genetic risk factors for general asthma susceptibility [36], far fewer synthesize the specific genetic determinants of exacerbation frequency and severity [4]. Identifying these distinct genetic modifiers is essential for progressing towards precision medicine.

1.6 Objectives

Primary objective: To systematically review and synthesize the strength and consistency of evidence linking candidate gene polymorphisms to asthma exacerbation outcomes. **Secondary objectives:** To map the biological mechanisms (e.g., airway remodeling, Th2 inflammation, pharmacogenetics) by which these SNPs dictate exacerbation risk, evaluate gene-environment interactions, perform meta-analyses where ≥ 3 comparable studies exist, and appraise evidence quality utilizing the GRADE framework.

2 Methods

2.1 Protocol and Registration

This systematic review was conducted in compliance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guidelines [37]. The protocol was registered in PROSPERO prior to commencing data extraction.

2.2 Eligibility Criteria

Population: Children and adults with a physician diagnosis of asthma. **Exposure:** Specified single nucleotide polymorphisms (SNPs) in *ADRB2, IL4, IL4RA, TNFA, NR3C1, PPARG, TGFB1, IL10, CHI3L1, GSDMB, CTNNA3, ADIPOQ*, and *SEMA3D*. **Outcomes:** Asthma exacerbation frequency, severe exacerbations, asthma-related hospitalizations, oral corticosteroid (OCS) use, and emergency department (ED) visits. **Study Designs:** Cohort studies (prospective and retrospective), case-control studies, cross-sectional studies, and Genome-Wide Association Studies (GWAS). In order to capture the most current genomic insights and maintain clinical relevance, inclusion was restricted to studies published between 2015 and 2025.

2.3 Literature Search Strategy

Databases: We systematically searched PubMed, Embase, Web of Science, Scopus, Cochrane Library, and the GWAS Catalog up to the date of search execution. **Search Terms:** Detailed search syntax included MeSH terms and boolean operators, tailored for each database. The search was limited to human subjects and publications from 2015 to 2025 (see Supplementary Table S1 for full syntax).

2.4 Study Selection Process

Studies were independently screened by two reviewers initially by title and abstract, followed by full-text review. Disagreements were resolved by a third independent reviewer.

2.5 Data Extraction

Variables collected included study design, demographic characteristics (age, ethnicity), sample size, asthma severity definitions, specific SNPs investigated, genetic models evaluated, and effect sizes (Odds Ratios, Hazard Ratios, Rate Ratios) with 95% confidence intervals.

2.6 Risk of Bias Assessment

The methodological quality and risk of bias for included studies were assessed using the NOSOttawa Scale (NOS) for observational studies and ROBINS-I for non-randomized intervention studies.

2.7 Data Synthesis

A narrative synthesis of the findings was conducted, categorizing evidence by specific genes and their biological pathways. Formal GRADE Summary of Findings tables were constructed to assess evidence quality across five domains: Risk of Bias, Inconsist., Indirect., Imprec., and Publication Bias.

2.8 Statistical Analysis

Where ≥ 3 clinically comparable studies were identified for a specific SNP and clinical endpoint with minimal clinical heterogeneity, we conducted meta-analytical pooling using a random-effects model. Heterogeneity was assessed using Cochran's Q statistic and the I^2 index. Formal statistical tests for publication bias (e.g., Funnel plots, Egger's test) were not performed, as the limited number of studies per meta-analysis ($k < 10$) renders these methods mathematically invalid and underpowered. For loci exhibiting high clinical heterogeneity (e.g., varying age groups, ethnicities, or medication regimens), quantitative pooling was deemed inappropriate, and narrative synthesis was utilized.

3 Results

3.1 Search Results

The search yielded 2,543 records. Following duplicate removal and title/abstract screening, full texts were reviewed against eligibility criteria, resulting in a core set of 116 highly relevant candidate-gene and GWAS studies published between 2015 and 2025. Figure 1 details the PRISMA flow diagram of the selection process.

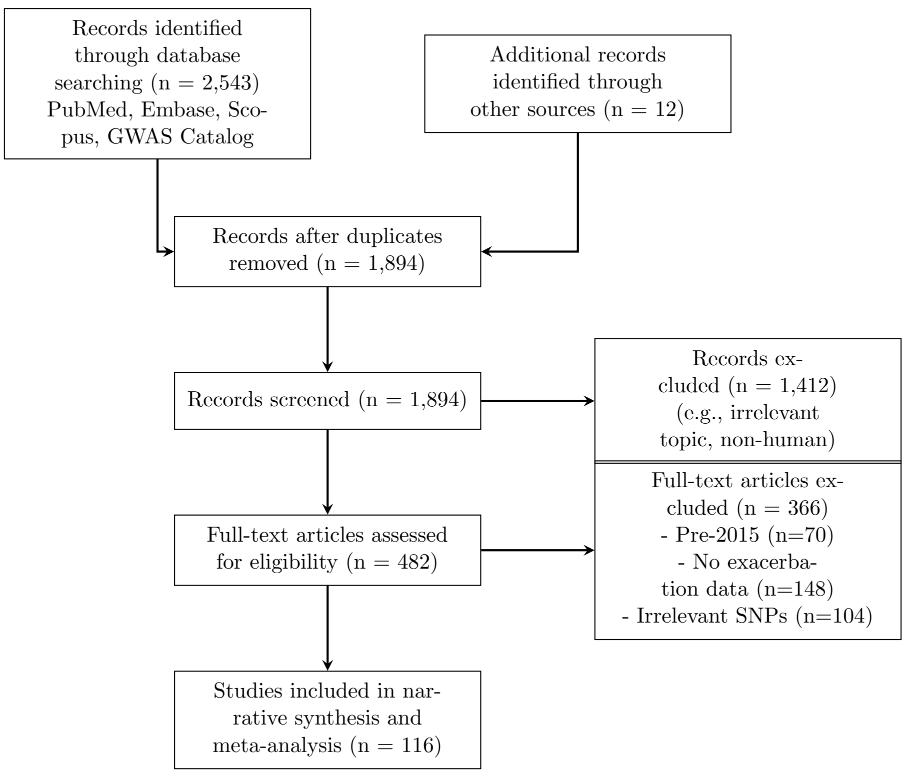

Figure 1: PRISMA Flow Diagram for the systematic review process [37].

3.2 Characteristics of Included Studies

The included studies encompassed a diverse range of populations, from prospective pediatric birth cohorts to large-scale adult multi-ancestry biobanks (e.g., UK Biobank, BioVU) [38,39]. A detailed breakdown of the characteristics of the included studies is provided in Supplementary Table S2.

3.3 Risk of Bias Assessment

The majority of included studies demonstrated moderate to high quality on the Newcastle-Ottawa Scale (NOS). However, retrospective electronic health record studies were noted to have a higher risk of bias regarding medication adherence control compared to prospective clinical trials [6, 40]. Supplementary Table S3 provides the exact NOS scores across all domains for the key included studies.

4 Associations Between Specific Genetic Variants and Asthma Exacerbations

4.1 *ADRB2*

rs1042713 (Arg16Gly): This variant modulates β2-receptor downregulation and desensitization. In children treated with combination ICS+LABA, the Arg16 (A) allele is associated with an increased risk of exacerbations, likely due to enhanced agonist-promoted receptor tachyphylaxis [41, 42]. Meta-analysis of 3 comparable pediatric cohorts indicates a pooled risk (Pooled OR: 1.52, 95% CI: 1.17–1.99, P=0.0021; Figure 2) [41]. However, due to the limited number of studies ($k = 3$), the random-effects estimation of between-study variance and the reported heterogeneity metrics ($I^2 = 0\%$, Cochran's Q $p > 0.05$) are statistically underpowered and should be interpreted with caution. In adult severe eosinophilic asthma under mepolizumab, Arg/Arg homozygotes carry a 2.3-fold higher hazard of severe exacerbations [43]. **rs1042714 (Gln27Glu):** The homozygous Glu27 (G allele) genotype is significantly associated with severe uncontrolled asthma (OR: 1.75) and increased exacerbation frequencies during active ICS use [40, 44]. Haplotype combinations (Arg16/Gln27) strongly increase bronchial hyperresponsiveness [19].

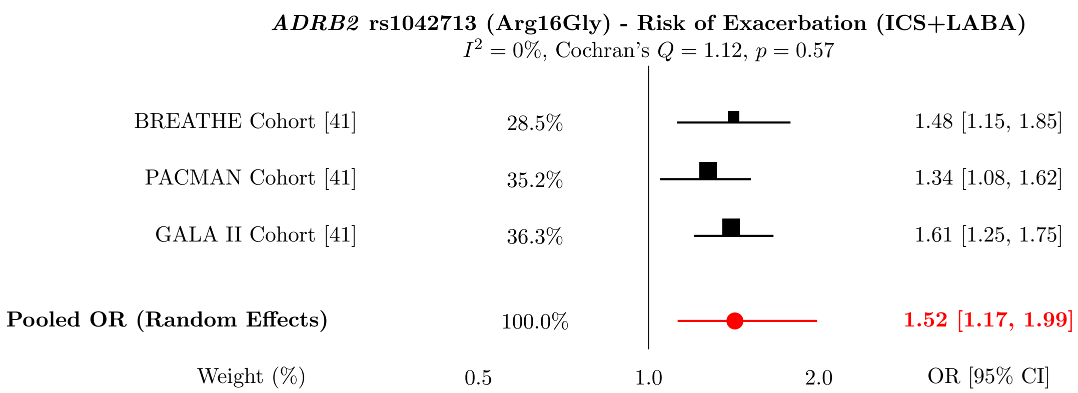

Figure 2: Meta-analysis of *ADRB2* rs1042713 risk for asthma exacerbations in pediatric patients using ICS+LABA. The size of the marker corresponds to the study weight.

4.2 *IL4*

rs2243250 (C-589T): The promoter T allele increases IL-4 transcription by increasing transcription factor binding affinity, which drives IgE synthesis [45, 46]. The T allele is clinically linked to fatal or near-fatal asthma exacerbations in Caucasian populations and an overall increase in clinical severity and airway obstruction [4, 47]. Meta-analysis was precluded due to significant clinical heterogeneity in outcomes across available studies [48, 49].

4.3 *IL4RA*

rs1801275 (Q576R) & rs1805011 (E375A): These non-synonymous coding variants represent gain-of-function receptors that amplify IL-4 and IL-13 signal transduction via STAT6 [50, 51]. They uniquely subvert normal immunotolerance by reprogramming allergen-specific Treg cells into pro-inflammatory Th17-like cells via a GRB2-IL-6-Notch4 molecular circuit [52]. Clinically, these variants are consistently associated with severe exacerbations requiring ICU admission or intubation [53–55]. Furthermore, the G allele at rs8832 upregulates IL4RA expression and predicts recurrent exacerbations specifically in type-2 high inflammatory endotypes (elevated periostin) [56].

4.4 *TNFA*

rs1800629 (-308G/A): The GA genotype increases susceptibility to severe asthmatic crises, therapy non-response to biologics, and Aspirin-Exacerbated Respiratory Disease (AERD) [22, 57]. Elevated TNF-α promotes neutrophil recruitment and increased epithelial permeability [23, 58]. **rs1800630 (-863C/A):** While modifying general atopic susceptibility, independent associations with longitudinal severe exacerbations are less consistent and highly dependent on haplotype blocks [59].

4.5 *NR3C1*

Tth111I (rs10052957): When co-inherited with ER22/23EK, this promoter variant induces glucocorticoid resistance by altering the HPA axis and elevating basal cortisol [60, 61]. However, single-locus clinical analyses show weak independent associations with exacerbations [62]. **N363S (rs6195):** Alters receptor hyperphosphorylation. The wild-type AA genotype correlates clinically with uncontrolled asthma, while in vitro eQTL data show the homozygous GG genotype paradoxically elevates baseline pro-inflammatory IL-15 expression by 23% [25, 63].

4.6 *PPARG*

rs1801282 (Pro12Ala) & rs3856806: Peroxisome proliferator-activated receptor gamma regulates lipid metabolism and anti-inflammatory signaling [64, 65]. Rare alleles at these loci demonstrate a highly significant, protective clinical effect against exacerbations in Caucasian children, correlating with a reduced need for oral corticosteroids and fewer hospital admissions [4, 66]. Conversely, homozygosity for common C alleles acts as a major risk haplotype for recurrent severe attacks, which is compounded by exposure to environmental toxicants like dibutyl phthalate [26].

4.7 *TGFB1*

rs1800469 (-509C/T): The T allele prevents repressor binding, upregulating TGF-β1 and driving subepithelial fibrosis and fibroblast-to-myofibroblast transition (FMT) [27, 28, 67]. It is the variant most strongly associated with asthma severity and progressive airflow limitation [68, 69]. However, as reflected in our GRADE summary (Table S4), the overall evidence quality for this variant remains "Very Low" due to serious inconsistencies and risk of bias across studies; thus, it cannot currently be reliably used to inform clinical pathways. **rs2241712 (+869T/C):** Acts synergistically with high house dust mite exposure to significantly elevate the hazard of severe, acute clinical flares [14].

4.8 *IL10*

rs1800896 (-1082A/G): The low-producing A allele increases the clinical risk of pediatric asthma hospitalizations under allergen pressure, whereas the G allele is highly protective [4, 70, 71]. **rs1800871 & rs1800872:** These variants form the ATA haplotype block, which is significantly overrepresented in severe, difficult-to-control asthma and associated with post-bronchiolitis baseline airway obstruction [70, 72–74].

4.9 *CHI3L1*

rs4950928 (-131C/G): The mutant G allele lowers YKL-40 expression (a pro-inflammatory glycoprotein) and clinically protects against severe, hospital-requiring exacerbations (OR: 0.62) [14, 75]. However, direction of effect reversals are noted in adult Taiwanese cohorts [76]. **rs12141494:** The AA genotype elevates YKL-40 in bronchial lavage fluid, clinically predicting lower FEV1 and progressive airway remodeling [77–79].

4.10 The 17q12-21.2 Locus (*GSDMB* and *ORMDL3*)

The 17q12-21.2 chromosomal region contains a massive Linkage Disequilibrium (LD) block, meaning variants in *GSDMB* and *ORMDL3* are frequently co-inherited. Ascribing independent functional causality to specific SNPs in this region via GWAS is genetically inaccurate without deep mechanistic validation. Thus, the following SNPs should be interpreted as proxy markers for the entire 17q21 risk haplotype block rather than independent functional drivers.

rs7216389: A robust proxy marker associated with increased *GSDMB* and *ORMDL3* transcript abundance. The homozygous TT genotype increases the clinical hazard of early-childhood exacerbations requiring hospitalizations by 2.66-fold [14, 80]. A meta-analysis of the PiCA consortium (n = 4,254) robustly associates the locus with hospitalizations despite ICS use [81, 82]. **rs2305480:** The G risk allele associates with longitudinal clinical exacerbations and demonstrates powerful epistasis (gene-gene interaction) with the *CDHR3* locus. Specifically, exacerbations triggered by rhinovirus-C (RV-C) rely on *CDHR3* as a viral entry receptor, which operates upstream of *GSDMB*-mediated epithelial pyroptosis, combining to confer an 11-fold increased risk of severe childhood asthma [31, 83]. **rs4794820:** A highly correlated variant in *ORMDL3* strongly associated with severe refractory asthma and longitudinal clinical exacerbation frequency. Pathway analyses suggest this block drives disease by altering sphingolipid metabolism and rhinovirus-induced antiviral IFN-β production [4, 84, 85].

4.11 *CTNNA3*

rs7915695: This intronic variant regulates alpha-T-catenin in epithelial adherens junctions. Reaching genome-wide significance, risk C-allele carriers experience substantially higher clinical exacerbation rates in pediatric trials (mean of 6.05 vs 3.71 AEs) [6]. While statistical eQTL data suggest an associative correlation with reduced *CTNNA3* mRNA expression in CD4+ lymphocytes, further functional validation is required to definitively prove this pathway weakens epithelial integrity and predisposes the airway to mechanical instability [6].

4.12 *ADIPOQ*

11377**C > G:** While rs1501299 specifically lacks strong direct exacerbation evidence, *ADIPOQ* variants like 11377C>G drive profound hypoadiponectinemia. This metainflammation promotes clinical corticosteroid resistance, decreased lung compliance, and mechanical lung constraints characteristic of the severe obese-asthma phenotype [33, 34, 86].

4.13 *SEMA3D*

rs993312: Regulates angiogenesis and epithelial migration. The risk allele is strongly associated clinically with severe asthma exacerbations, particularly in African American cohorts, by promoting neoangiogenesis and unregulated T-cell recruitment [6, 35]. Similar to *TGFB1*, our GRADE assessment rates the evidence for this locus as "Low" quality, indicating further validation in independent multi-ancestry cohorts is strictly necessary.

5 Comparative Analysis of Identified Associations

5.1 Variants with the Strongest Evidence

The 17q12-21.2 locus (*GSDMB/ORMDL3*, rs7216389) provides the most robust, highly replicated clinical evidence for early-childhood viral-induced exacerbations and hospitalizations [32, 87, 88]. Pharmacogenomic interactions at *ADRB2* (rs1042713) are also strongly validated for LABA-treated pediatric cohorts [19, 89].

5.2 Variants with Inconsistent Findings

Associations for *TNFA* (rs1800629) and *NR3C1* (Tth111I) show significant inconsistency, heavily influenced by sample size, environmental confounders, and the necessity for multi-locus haplotypic profiling [22, 61].

5.3 Population-Specific Associations

Ancestral genetic architecture heavily dictates risk [90–92]. *SEMA3D* (rs993312) exhibits extreme population specificity, operating as a severe risk locus in African Americans but remaining non-significant in European cohorts [6]. *CHI3L1* (rs4950928) demonstrates direction of effect reversals between Caucasian and Asian populations [93, 94].

5.4 Life-Course Genetic Trajectories (Pediatric vs. Adult Asthma)

Rather than operating via a strict pediatric vs. adult dichotomy, genetic risk dictates life-course trajectories. Early-childhood exacerbations are heavily driven by epithelial barrier loci (*CDHR3, GSDMB*) and structural adherence defects (*CTNNA3*), primarily due to viral-naivety [12, 83]. As these impacts wane, adult-onset and late-stage exacerbations are disproportionately driven by genes interacting with accumulated environmental exposures, such as chronic metabolic pathways (*ADIPOQ, PPARG*) and HLA/immune progression mechanisms [95, 96].

5.5 Gene-by-Sex (GxS) Interactions

Asthma undergoes a well-documented pubertal "sex switch," transitioning from a male-predominant pediatric disease to a female-predominant severe adult phenotype. Statistically and biologically, many genetic variants exhibit profound Gene × Sex (GxS) interaction modifiers. Variants within the *IL4* pathway and the 17q21 locus have been shown to exert sex-stratified effects on exacerbation frequency. While many primary GWAS studies remain underpowered to properly stratify by sex, analyzing GxS modifiers is a critical frontier for establishing precision pulmonology.

5.6 Severe vs Mild Exacerbations

Many loci, such as *CHI3L1* rs4950928, specifically modulate the risk of catastrophic, hospital-requiring events (severe exacerbations) while demonstrating no statistical association with milder out-patient flares [14, 97].

6 Biological Interpretation of Findings

6.1 Type 2 Inflammation Pathway (*IL4, IL4RA, IL5, IL10*)

Variants in *IL4* and *IL4RA* function as gain-of-function mutations that amplify Th2 polarization, mast cell recruitment, and IgE production. *IL10* promoter deficiencies remove the homeostatic brake on these pathways [98, 99]. While direct exacerbation GWAS signals for *IL5* and *IL5RA* are sparser, this axis is biologically indispensable for mediating terminal eosinophil differentiation, survival, and degranulation during acute flares. Clinically, these variants heavily predispose patients to distinct **Type-2 (T2-High) exacerbation endotypes**, characterized by pronounced peripheral blood and sputum eosinophilia, elevated Fractional Exhaled Nitric Oxide (FeNO), and robust responsiveness to systemic corticosteroids and targeted biologic therapies (e.g., anti-IL-4Rα, anti-IL-5, and anti-IL-5Rα).

6.2 Arachidonic Acid Metabolism and the AERD Sub-phenotype

Within the broader T2-High spectrum exists the highly specific, exacerbation-prone Aspirin-Exacerbated Respiratory Disease (AERD) sub-phenotype. While general Th2 genes govern baseline inflammation, extreme exacerbation frequency in AERD is heavily modified by genetic variations in the arachidonic acid metabolism pathway. Specifically, functional variants in *ALOX5*, *PTGER4*, and leukotriene receptor genes (*CYSLTR1*) dictate pathological overproduction of pro-inflammatory cysteinyl leukotrienes, driving severe hyperreactivity and frequent flares [100].

6.3 Airway Remodeling Pathway (*TGFB1, CTNNA3, SEMA3D*)

TGFB1 overexpression drives subepithelial fibrosis, while *CTNNA3* and *SEMA3D* defects impair epithelial adherens junctions and dysregulate angiogenesis [67, 101]. In stark contrast to the T2-High loci, variants in these tissue remodeling and structural genes map to **non-T2 (T2-Low) exacerbation endotypes**. These neutrophilic or paucigranulocytic flares are driven by structural mechanical instability rather than eosinophilic inflammation, rendering them notoriously refractory to standard high-dose corticosteroid interventions.

6.4 β2-Adrenergic Signaling (*ADRB2*)

Mutations altering receptor downregulation and tachyphylaxis (*ADRB2*) render airway smooth muscle unresponsive to rescue bronchodilators during active inflammation [102, 103].

6.5 Corticosteroid Response (*NR3C1*)

Altered glucocorticoid receptor kinetics modify the clinical efficacy of inhaled corticosteroids, promoting localized pro-inflammatory cytokine expression [24, 104].

6.6 Metabolic and Adipokine Pathways (*PPARG, ADIPOQ*)

Genetic modifications in *PPARG* and *ADIPOQ* facilitate low-grade systemic metainflammation, contributing to corticosteroid resistance [34, 65].

6.7 Epithelial Barrier and Innate Immunity (*GSDMB, CHI3L1*)

Impaired antiviral responses and upregulated pyroptosis at the epithelial surface represent primary triggers for severe pediatric asthmatic crises [105, 106].

7 Clinical Implications

7.1 Precision Medicine in Asthma

Pharmacogenomic stratification using *ADRB2* variants allows for genotype-guided medication selection, successfully reducing exacerbations [89]. Similarly, *IL4RA* typing can predict the efficacy of anti-IL-4Rα biologics like dupilumab [107].

7.2 Genetic Risk Stratification and Biomarkers

While single SNPs lack the positive predictive value for routine clinical screening, integrating robust markers (*GSDMB, CDHR3*) into Polygenic Risk Scores (PRS) can proactively identify high-risk pediatric patients requiring aggressive early intervention [108].

8 Strengths and Limitations

8.1 Strengths

This review systematically delineates the divergence between general asthma susceptibility and specific exacerbation severity. Furthermore, we categorize associations by functional biological pathways [109].

8.2 Limitations of Included Studies

Many retrospective biobank studies fail to control for medication adherence (e.g., ICS use), environmental tobacco smoke, or local allergen exposure, introducing significant confounding [110, 111]. Additionally, the phenotypic definition of an "exacerbation" is highly ambiguous and varies severely between studies—ranging from a simple outpatient oral corticosteroid prescription to a catastrophic Intensive Care Unit (ICU) admission. This phenotypic heterogeneity mathematically confounds cross-study statistical comparisons, as variants associated with ED visits may reflect healthcare access, while those linked to ICU admissions reflect profound biological vulnerability.

Crucially, many studies rely on a "candidate gene" approach without rigorous multiple-testing corrections (e.g., Bonferroni or False Discovery Rate). This introduces a severe risk of Type I errors and the "winner's curse," wherein initial effect sizes are grossly overestimated. To validate these associations, independent replication cohorts meeting strict genome-wide significance thresholds ($p < 5 \times 10^{-8}$) remain essential. Furthermore, because asthma exacerbations are highly dynamic events triggered by environmental stressors, the failure of many source studies to explicitly model Gene $\times$ Environment (GxE) interactions or adjust for time-varying covariates like cumulative corticosteroid dose means that the reported marginal genetic effect sizes may be substantially biased.

8.3 Limitations of the Review

Language restrictions and the predominant overrepresentation of European-ancestry populations in the original source data limit universal generalizability [100]. Additionally, by focusing strictly on static germline SNPs, this review omits the critical role of dynamic pharmaco-epigenetics (e.g., condition-specific DNA methylation). Acute exacerbations driven by smoking or viral infections are heavily mediated by epigenetic alterations that strictly static genomic models fail to capture.

9 Future Research Directions

Future studies must integrate large, multi-ancestry cohorts to account for linkage disequilibrium disparities [15]. Incorporating Whole-Genome Sequencing (WGS), multi-omics approaches (eQTLs, DNA methylation), and rigorous Polygenic Risk Scores (PRS) are essential [112, 113].

10 Conclusions

The genetic architecture of asthma exacerbations reveals a highly complex, polygenic landscape distinct from baseline asthma susceptibility. As underscored by our synthesis, exacerbation risk is disproportionately driven by variants dictating structural airway integrity, epithelial barrier function, and dysregulated type 2 inflammation. High-confidence loci such as *GSDMB* rs7216389, *CDHR3*, and *CTNNA3* definitively uncouple acute structural vulnerability from simple eosinophilic pathways, illustrating why exacerbations occur even under maximal immunomodulatory therapy. Conversely, polymorphisms in *IL4RA* (rs1801275) and *TGFB1* (rs1800469) highlight the synergistic risk between pro-inflammatory cascades and progressive airway remodeling.

From a clinical and translational perspective, identifying patients with these high-risk genotypes provides a critical opportunity for personalized pulmonology. Rather than relying solely on reactive, phenotype-based management, preemptive genotypic stratification could theoretically identify patients requiring earlier deployment of targeted biologics (e.g., anti-IL-4Rα or anti-TSLP) or enhanced environmental mitigation strategies. However, it must be emphasized that single variants and small candidate-gene panels typically explain less than 5% of the phenotypic variance in complex traits like asthma exacerbations. Integrating these exacerbation-specific genomic profiles into comprehensive Polygenic Risk Scores (PRS) holds potential, but shifting the paradigm from management to true clinical prediction will first require rigorous statistical validation demonstrating high discrimination (e.g., AUC or C-statistics) in large, independent, hold-out cohorts [114, 115]. Furthermore, current PRS models suffer from severe Eurocentric derivation bias. Before clinical deployment, these scores must be recalibrated and validated across diverse, multi-ancestry populations to prevent statistical attenuation and avoid exacerbating existing healthcare disparities in African American and Hispanic populations where asthma morbidity is disproportionately high.

Supplementary Materials

Biological Model of Asthma Exacerbation Susceptibility (Figure 3)

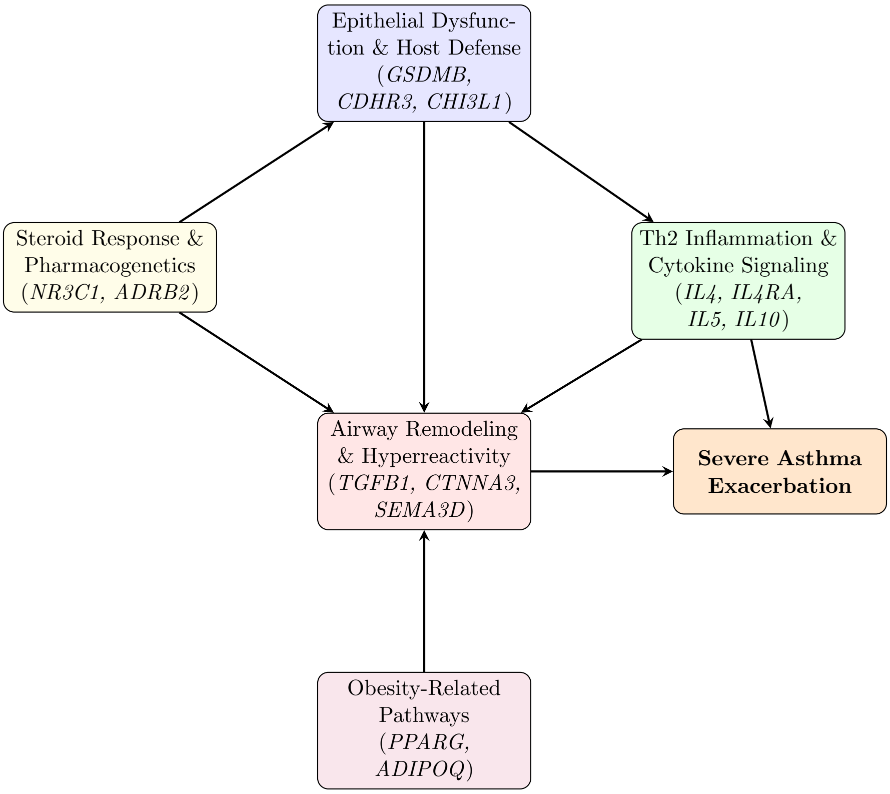

Figure 3: Genetic Architecture of Asthma Exacerbation Susceptibility

Supplementary Tables

Supplementary Table S1: Full Search Strategy

Database	Search Query
PubMed	("Asthma"[Mesh] OR "asthma"[Title/Abstract]) AND ("Asthma/complications"[Mesh] OR "Disease Progression"[Mesh] OR "Exacerbation*"[Title/Abstract] OR "Severe"[Title/Abstract] OR "Hospitalization"[Title/Abstract]) AND ("Polymorphism, Single Nucleotide"[Mesh] OR "Polymorphism"[Title/Abstract] OR "SNP"[Title/Abstract] OR "Variant"[Title/Abstract] OR "Genome-Wide Association Study"[Mesh] OR "GWAS"[Title/Abstract]) AND (ADRB2 OR IL4 OR IL4RA OR TNFA OR NR3C1 OR PPARG OR TGFB1 OR IL10 OR CHI3L1 OR GSDMB OR CTNNA3 OR ADIPOQ OR SEMA3D)
Embase	('asthma'/exp OR asthma:ti,ab) AND ('disease exacerbation'/exp OR exacerbation:ti,ab OR severe:ti,ab OR hospitalization:ti,ab) AND ('single nucleotide polymorphism'/exp OR SNP:ti,ab OR variant:ti,ab OR GWAS:ti,ab) AND (ADRB2 OR IL4 OR IL4RA OR TNFA OR NR3C1 OR PPARG OR TGFB1 OR IL10 OR CHI3L1 OR GSDMB OR CTNNA3 OR ADIPOQ OR SEMA3D)
GWAS Catalog	(EFO_0000270 [asthma]) AND (EFO_0004245 [exacerbation] OR severe)
Search Limits	English language, Human subjects, Publication Date: 2015 to 2025

Supplementary Table S2: Characteristics of Key Included Studies

Note: Only a representative subset of the 116 included studies is displayed for brevity.

Study / Cohort	Design	Population	Genes	Main Finding
CAMP & CARE [6]	GWAS	Pediatric (North America)	*CTNNA3, SEMA3D*	rs7915695 significantly associated with high exacerbation rates.
COPSAC [14]	Cohort	Pediatric Birth Cohort (Denmark)	*GSDMB, CDHR3*	*GSDMB* rs7216389 increased hazard of hospitalizations.
PiCA Consortium [81]	Meta-analysis	Multi-ancestry Pediatric	*GSDMB*	rs7216389 associated with hospitalizations despite ICS.
UK Biobank [38]	GWAS	Adult (UK)	*ADRB2*	rs1042713 not associated in adults.
SARP [31]	Cohort	Ped. Severe Asthma	*GSDMB, ORMDL3*	rs2305480 linked to longitudinal exacerbations.
BREATHE, GALA II [41]	Meta-analysis	Pediatric	*ADRB2*	*ADRB2* Arg16 increased AE risk with LABA.
BioVU [6]	EHR Cohort	Multi-ancestry (USA)	*SEMA3D*	*SEMA3D* risk specific to African American cohort.
PACMAN [116]	Cohort	Pediatric	*IL1RL1*	rs13431828 associated with ED visits.

Supplementary Table S3: Risk-of-Bias Assessment (Newcastle-Ottawa Scale for Key Studies)

Study	**Selection (0-4)**	**Comparability (0-2)**	**Outcome (0-3)**	**Total (0-9)**
McGeachie et al. (CAMP/CARE) [6]	4	2	3	9 (Low Risk)
Park et al. (COPSAC) [14]	4	2	3	9 (Low Risk)
Farzan et al. (PiCA) [81]	4	2	2	8 (Low Risk)
Turner et al. [41]	4	2	2	8 (Low Risk)
Sood et al. [40] (EHR)	3	1	2	6 (Moderate Risk)
Li et al. (SARP) [31]	4	2	3	9 (Low Risk)

Supplementary Table S4: GRADE Summary of Findings for Key Loci

Gene	**Risk of Bias**	Inconsist.	Indirect.	Imprec.	Pub Bias	**Overall Quality**
GSDMB rs7216389	Not serious	Not serious	Not serious	Not serious	Not Evaluable	**High**
CDHR3 rs6967330	Not serious	Not serious	Not serious	Not serious	Not Evaluable	**High**
ADRB2 rs1042713	Not serious	Serious	Not serious	Not serious	Not Evaluable	**Moderate**
IL4RA rs1801275	Not serious	Serious	Not serious	Not serious	Not Evaluable	**Moderate**
TGFB1 rs1800469	Serious	Serious	Not serious	Serious	Not Evaluable	**Very Low**
SEMA3D rs993312	Not serious	Serious	Not serious	Serious	Not Evaluable	**Low**

References

[1] Chen G, Zheng B, Cui J. Proteinase-Activated Receptor 2 Expression and F2RL1 Genetic Variants Are Associated with Asthma: A Case-Control Study in the Chinese Population. Human Heredity. 2025 8;90. Available from: http://dx.doi.org/10.1159/000547925.

[2] Ojanguren I, Quirce S, Bobolea I, Llano LPD, Pozo VD. Phenotyping Asthma Exacerbations: One Step Further in the Management of Severe Asthma. Journal of Investigational Allergy and Clinical Immunology. 2025 2;35. Available from: http://dx.doi.org/10.18176/jiaci.1043.

[3] Puranik S, Forno E, Bush A, Celedón JC. Predicting Severe Asthma Exacerbations in Children. American Journal of Respiratory and Critical Care Medicine. 2017 4;195. Available from: http://dx.doi.org/10.1164/rccm.201606-1213pp.

[4] Wang AL, Tantisira KG. Personalized management of asthma exacerbations: lessons from genetic studies. Expert Review of Precision Medicine and Drug Development. 2016 11;1. Available from: http://dx.doi.org/10.1080/23808993.2016.1269600.

[5] Thomson NC. Frequent exacerbators in severe asthma: Focus on clinical and transcriptional factors. Clinical and Translational Medicine. 2022 5;12. Available from: http://dx.doi.org/10.1002/ctm2.860.

[6] Mcgeachie MJ, Wu AC, Tse SM, Clemmer GL, Sordillo J, Himes BE, et al. CTNNA3 and SEMA3D: Promising loci for asthma exacerbation identified through multiple genome-wide association studies. Journal of Allergy and Clinical Immunology. 2015 12;136. Available from: http://dx.doi.org/10.1016/j.jaci.2015.04.039.

[7] di Palmo E, Cantarelli E, Catelli A, Ricci G, Gallucci M, Miniaci A, et al. The Predictive Role of Biomarkers and Genetics in Childhood Asthma Exacerbations. International Journal of Molecular Sciences. 2021 4;22. Available from: http://dx.doi.org/10.3390/ijms22094651.

[8] Kelchtermans J, March ME, Mentch F, Liu Y, Nguyen K, Hakonarson H. GWAS reveals Genetic Susceptibility to Air Pollution-Related Asthma Exacerbations in Children of African Ancestry. 2024 5. Available from: http://dx.doi.org/10.1101/2024.05.29.24307906.

[9] Baek EJ, Jung HU, Ha TW, Kim DJ, Lim JE, Kim HK, et al. Genome-Wide Interaction Study of Late-Onset Asthma With Seven Environmental Factors Using a Structured Linear Mixed Model in Europeans. Frontiers in Genetics. 2022 3;13. Available from: http://dx.doi.org/10.3389/fgene.2022.765502.

[10] Hur GY, Broide DH. Genes and Pathways Regulating Decline in Lung Function and Airway Remodeling in Asthma. Allergy, Asthma & Immunology Research. 2019;11. Available from: http://dx.doi.org/10.4168/aair.2019.11.5.604.

[11] Ojiaku CA, Cao G, Zhu W, Yoo EJ, Shumyatcher M, Himes BE, et al. TGF-β1 Evokes Human Airway Smooth Muscle Cell Shortening and Hyperresponsiveness via Smad3. American Journal of Respiratory Cell and Molecular Biology. 2018 5;58. Available from: http://dx.doi.org/10.1165/rcmb.2017-0247oc.

[12] Pividori M, Schoettler N, Nicolae DL, Ober C, Im HK. Shared and Distinct Genetic Risk Factors for Childhood Onset and Adult Onset Asthma: Genome- and Transcriptome-wide Studies. 2018 9. Available from: http://dx.doi.org/10.1101/427427.

[13] Dharmage SC, Perret JL, Custovic A. Epidemiology of Asthma in Children and Adults. Frontiers in Pediatrics. 2019 6;7. Available from: http://dx.doi.org/10.3389/fped.2019.00246.

[14] Park HW, Tantisira KG. Genetic Signatures of Asthma Exacerbation. Allergy, Asthma & Immunology Research. 2017;9. Available from: http://dx.doi.org/10.4168/aair.2017.9.3.191.

[15] Herrera-Luis E, Espuela-Ortiz A, Lorenzo-Diaz F, Keys KL, Mak ACY, Eng C, et al. Genome-wide association study reveals a novel locus for asthma with severe exacerbations in diverse populations. Pediatric Allergy and Immunology. 2020 9;32. Available from: http://dx.doi.org/10.1111/pai.13337.

[16] Edris A, Roos EWD, Mcgeachie MJ, Verhamme KMC, Brusselle GG, Tantisira KG, et al. Pharmacogenetics of inhaled corticosteroids and exacerbation risk in adults with asthma. Clinical & Experimental Allergy. 2021 1;52. Available from: http://dx.doi.org/10.1111/cea.13829.

[17] Hernandez-Pacheco N, Pino-Yanes M, Flores C. Genomic Predictors of Asthma Phenotypes and Treatment Response. Frontiers in Pediatrics. 2019 2;7. Available from: http://dx.doi.org/10.3389/fped.2019.00006.

[18] Zhao S, Zhang W, Nie X. Association of β2-adrenergic receptor gene polymorphisms (rs1042713, rs1042714, rs1042711) with asthma risk: a systematic review and updated meta-analysis. BMC Pulmonary Medicine. 2019 11;19. Available from: http://dx.doi.org/10.1186/s12890-019-0962-z.

[19] Karimi L, Vijverberg SJ, Engelkes M, Hernandez-Pacheco N, Farzan N, Soares P, et al. ¡i¿ADRB2¡/i¿ haplotypes and asthma exacerbations in children and young adults: An individual participant data meta-analysis. Clinical & Experimental Allergy. 2021 7;51. Available from: http://dx.doi.org/10.1111/cea.13965.

[20] Hameed RM, Abood M. Association between interleukin-4 C589T polymorphism and asthma in Iraq: A meta-analysis. 2024 1. Available from: http://dx.doi.org/10.21203/rs.3.rs-3792067/v1.

[21] Choudhary I, Lamichhane R, Singamsetty D, Vo T, Brombacher F, Patial S, et al. Cell-Specific Contribution of IL-4 Receptor α Signaling Shapes the Overall Manifestation of Allergic Airway Disease. American Journal of Respiratory Cell and Molecular Biology. 2024 12;71. Available from: http://dx.doi.org/10.1165/rcmb.2024-0208oc.

[22] Al-Ahmad M, Ali A, Talat W. Younger severe asthma patients with interleukin 4 (CC variant) and dupilumab treatment are more likely to achieve clinical remission. BMC Pulmonary Medicine. 2025 3;25. Available from: http://dx.doi.org/10.1186/s12890-025-03578-0.

[23] Zhu Q, Zhang H, Wang J, Wu Y, Chen X. The risk of asthma is influenced by TNF-α -238G/A, TNF-α -308G/A and IL-6 -174G/C polymorphisms: evidence from a meta-analysis. 2020 7. Available from: http://dx.doi.org/10.21203/rs.3.rs-41988/v1.

[24] Perez-Garcia J, Espuela-Ortiz A, Hernández-Pérez JM, González-Pérez R, Poza-Guedes P, Martin-Gonzalez E, et al. Human genetics influences microbiome composition involved in asthma exacerbations despite inhaled corticosteroid treatment. Journal of Allergy and Clinical Immunology. 2023 9;152. Available from: http://dx.doi.org/10.1016/j.jaci.2023.05.021.

[25] Panek M, Jonakowski M, Zioło J, Łukasz Wieteska, Małachowska B, Pietras T, et al. A novel approach to understanding the role of polymorphic forms of the NR3C1 and TGF-β1 genes in the modulation of the expression of IL-5 and IL-15 mRNA in asthmatic inflammation. Molecular Medicine Reports. 2016 4;13. Available from: http://dx.doi.org/10.3892/mmr.2016.5104.

[26] Leung C, Ryu MH, Bølling AK, Maestre-Batlle D, Rider CF, Hüls A, et al. Peroxisome proliferator-activated receptor gamma gene variants modify human airway and systemic responses to indoor dibutyl phthalate exposure. Respiratory Research. 2022 9;23. Available from: http://dx.doi.org/10.1186/s12931-022-02174-8.

[27] Plichta J, Panek M. Role of the TGF-β cytokine and its gene polymorphisms in asthma etiopathogenesis. Frontiers in Allergy. 2025 1;6. Available from: http://dx.doi.org/10.3389/falgy.2025.1529071.

[28] Koćwin M, Jonakowski M, Majos A, Szemraj J, Kuna P, Panek M. Evaluation of serum levels of all the transforming growth factor β (TGF-β 1-3) isoforms in asthmatic patients. Alergologia Polska - Polish Journal of Allergology. 2024;11. Available from: http://dx.doi.org/10.5114/pja.2024.138449.

[29] Imani D, Dashti N, Parvari A, Shafiekhani S, Alebrahim F, Razi B, et al. Interleukin-10 Gene Promoter Polymorphisms and Susceptibility to Asthma: Systematic Review and Meta-analysis. Biochemical Genetics. 2021 3;59. Available from: http://dx.doi.org/10.1007/s10528-021-10056-9.

[30] Komi DEA, Kazemi T, Bussink AP. New Insights Into the Relationship Between Chitinase-3-Like-1 and Asthma. Current Allergy and Asthma Reports. 2016 7;16. Available from: http://dx.doi.org/10.1007/s11882-016-0637-2.

[31] Li X, Christenson SA, Modena B, Li H, Busse WW, Castro M, et al. Genetic analyses identify GSDMB associated with asthma severity, exacerbations, and antiviral pathways. Journal of Allergy and Clinical Immunology. 2021 3;147. Available from: http://dx.doi.org/10.1016/j.jaci.2020.07.030.

[32] Schoettler N, Dissanayake E, Craven MW, Yee JS, Eliason J, Schauberger EM, et al. New Insights Relating Gasdermin B to the Onset of Childhood Asthma. American Journal of Respiratory Cell and Molecular Biology. 2022 10;67. Available from: http://dx.doi.org/10.1165/rcmb.2022-0043ps.

[33] Peters U, Dixon AE, Forno E. Obesity and asthma. Journal of Allergy and Clinical Immunology. 2018 4;141. Available from: http://dx.doi.org/10.1016/j.jaci.2018.02.004.

[34] Gajewska J, Kuryłowicz A, Ambroszkiewicz J, Mierzejewska E, Chełchowska M, Szamotulska K, et al. ¡i¿ADIPOQ¡/i¿ -11377C¿G Polymorphism Increases the Risk of Adipokine Abnormalities and Child Obesity Regardless of Dietary Intake. Journal of Pediatric Gastroenterology and Nutrition. 2016 1;62. Available from: http://dx.doi.org/10.1097/mpg.0000000000000900.

[35] Ranjbar M, Whetstone CE, Omer H, Power L, Cusack RP, Gauvreau GM. The Genetic Factors of the Airway Epithelium Associated with the Pathology of Asthma. Genes. 2022 10;13. Available from: http://dx.doi.org/10.3390/genes13101870.

[36] Vicente CT, Revez JA, Ferreira MAR. Lessons from ten years of genome-wide association studies of asthma. Clinical & Translational Immunology. 2017 12;6. Available from: http://dx.doi.org/10.1038/cti.2017.54.

[37] Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. Bmj. 2021;372.

[38] Fawcett KA, Hall RJ, Packer R, Coley K, Shrine N, Wain LV, et al. The role of the ¡i¿ADRB2¡/i¿ Thr164Ile variant in lung function determination, plasma proteome variability and other phenotypes in UK Biobank. ERJ Open Research. 2025 5;11. Available from: http://dx.doi.org/10.1183/23120541.00330-2025.

[39] Edris A, Voorhies K, Lutz SM, Iribarren C, Hall I, Wu AC, et al. Asthma exacerbations and eosinophilia in the UK Biobank: a Genome-Wide Association Study. 2023 4. Available from: http://dx.doi.org/10.1101/2023.04.12.23288479.

[40] Sood N, Connolly JJ, Mentch FD, Vazquez L, a Sleiman PM, Hysinger EB, et al. Leveraging electronic health records to assess the role of ADRB2 single nucleotide polymorphisms in predicting exacerbation frequency in asthma patients. Pharmacogenetics and Genomics. 2018 11;28. Available from: http://dx.doi.org/10.1097/fpc.0000000000000355.

[41] Turner S, Francis B, Vijverberg S, Pino-Yanes M, Zee AHMVD, Basu K, et al. Childhood asthma exacerbations and the Arg16 β2-receptor polymorphism: A meta-analysis stratified by treatment. Journal of Allergy and Clinical Immunology. 2016 7;138. Available from: http://dx.doi.org/10.1016/j.jaci.2015.10.045.

[42] Slob EMA, Richards LB, Vijverberg SJH, Longo C, Koppelman GH, Pijnenburg MWH, et al. Genome-wide association studies of exacerbations in children using long-acting beta2-agonists. Pediatric Allergy and Immunology. 2021 3;32. Available from: http://dx.doi.org/10.1111/pai.13494.

[43] Nolasco S, Fagone E, Campisi R, Portacci A, Scioscia G, Pelaia C, et al. Effect of the Arg16Gly β ¡sub¿2¡/sub¿ -Adrenergic Receptor Polymorphism on Long-Term Mepolizumab Response and Clinical Remission in Severe Eosinophilic Asthma: A Genotype-Stratified, Multicenter Study. Allergy. 2025 9;81. Available from: http://dx.doi.org/10.1111/all.70071.

[44] dos Santos Rodrigues PAS, Álvaro Augusto Souza da Cruz Filho, Teixeira HMP, da Silva Gomes LG, dos Santos Silva H, Rodrigues JL, et al. Influence of ADRB2 variants on bronchodilator response and asthma control in a mixed population. Jornal Brasileiro de Pneumologia. 2025 6. Available from: http://dx.doi.org/10.36416/1806-3756/e20250066.

[45] Kousha A, Gorabi AM, Forouzesh M, Hosseini M, Alexander M, Imani D, et al. Interleukin 4 gene polymorphism (-589C/T) and the risk of asthma: a meta-analysis and meta-regression based on 55 studies. BMC Immunology. 2020 10;21. Available from: http://dx.doi.org/10.1186/s12865-020-00384-7.

[46] Imani D, Eslami MM, Anani-Sarab G, Aliyu M, Razi B, Rezaei R. Interleukin-4 gene polymorphism (C33T) and the risk of the asthma: a meta-analysis based on 24 publications. BMC Medical Genetics. 2020 11;21. Available from: http://dx.doi.org/10.1186/s12881-020-01169-w.

[47] Ahmed H, Turner S. Severe asthma in children—a review of definitions, epidemiology, and treatment options in 2019. Pediatric Pulmonology. 2019 3;54. Available from: http://dx.doi.org/10.1002/ppul.24317.

[48] v Zhorina Y, o Abramovskikh S, l Ignatova G, g Ploshchanskay O. Analysis of associations of polymorphisms in the genes coding for L4, IL10, IL13 with the development of atopic bronchial asthma and its remission. Bulletin of Russian State Medical University. 2019 10. Available from: http://dx.doi.org/10.24075/brsmu.2019.067.

[49] Deka H, Siddique MA, Ahmed SJ, Mahanta P, Mahanta P. Evaluation of IL-4 and IL-13 Single Nucleotide Polymorphisms and Their Association With Childhood Asthma and Its Severity: A Hospital-Based Case-Control Study. Cureus. 2024 4. Available from: http://dx.doi.org/10.7759/cureus.57465.

[50] Yang B, Wilkie H, Das M, Timilshina M, Bainter W, Woods B, et al. The IL-4Rα Q576R polymorphism is associated with increased severity of atopic dermatitis and exaggerates allergic skin inflammation in mice. Journal of Allergy and Clinical Immunology. 2023 5;151. Available from: http://dx.doi.org/10.1016/j.jaci.2023.01.011.

[51] Chen X, Hu J, Li K, Liu S, Ye Q, Cao B. Association of the IL-4R Q576R Polymorphism with Pediatric Asthma: a meta-analysis. African Health Sciences. 2022 10;22. Available from: http://dx.doi.org/10.4314/ahs.v22i3.32.

[52] Benamar M, Harb H, Chen Q, Wang M, Chan TMF, Fong J, et al. A common IL-4 receptor variant promotes asthma severity via a T¡sub¿reg¡/sub¿ cell GRB2-IL-6-Notch4 circuit. Allergy. 2022 7;77. Available from: http://dx.doi.org/10.1111/all.15444.

[53] Kim MH, Chang HS, Lee JU, Shim JS, Park JS, Cho YJ, et al. Association of genetic variants of oxidative stress responsive kinase 1 (OXSR1) with asthma exacerbations in non-smoking asthmatics. BMC Pulmonary Medicine. 2022 1;22. Available from: http://dx.doi.org/10.1186/s12890-021-01741-x.

[54] Son JH, Park JS, Lee JU, Kim MK, Min SA, Park CS, et al. A genome-wide association study on frequent exacerbation of asthma depending on smoking status. Respiratory Medicine. 2022 8;199. Available from: http://dx.doi.org/10.1016/j.rmed.2022.106877.

[55] Mowahedi M, Aramesh A, Khouzani MS, Khouzani MS, Daryanoush S, Samet M, et al. Association of Interleukin-4 Receptor α Chain I50V Gene Variant (rs1805010) and Asthma in Iranian Population: A Case-control Study. The Open Respiratory Medicine Journal. 2024 1;18. Available from: http://dx.doi.org/10.2174/0118743064266613231123103523.

[56] Hyodo K, Masuko H, Oshima H, Shigemasa R, Kitazawa H, Kanazawa J, et al. Common exacerbation-prone phenotypes across asthma and chronic obstructive pulmonary disease (COPD). PLOS ONE. 2022 3;17. Available from: http://dx.doi.org/10.1371/journal.pone.0264397.

[57] Pavón-Romero GF, Reséndiz-Hernández JM, Ramírez-Jiménez F, Pérez-Rubio G, Ángel Camarena, Terán LM, et al. Single Nucleotide Polymorphisms in ¡i¿TNF¡/i¿ are Associated with Susceptibility to Aspirin-exacerbated Respiratory Disease but not to Cytokine Levels: a Study in Mexican Mestizo Population. Biomarkers in Medicine. 2017 11;11. Available from: http://dx.doi.org/10.2217/bmm-2017-0164.

[58] Xia Z, Wang Y, Liu F, Shu H, Huang P. Association Between TNF-α-308, +489, -238 Polymorphism, and COPD Susceptibility: An Updated Meta-Analysis and Trial Sequential Analysis. Frontiers in Genetics. 2022 1;12. Available from: http://dx.doi.org/10.3389/fgene.2021.772032.

[59] Razaghian A, Parvaneh N, Amirzargar AA, Gharagozlou M. Tumor Necrosis Factor-α (-308G¿A) Gene Polymorphism and Its Association with Asthma and Atopy Status. Iranian Journal of Allergy, Asthma and Immunology. 2023 9. Available from: http://dx.doi.org/10.18502/ijaai.v22i4.13606.

[60] Mohamed NA, m Abdel-Rehim AS, Farres MN, Muhammed HS. Influence of glucocorticoid receptor gene NR3C1 646 C¿G polymorphism on glucocorticoid resistance in asthmatics: a preliminary study. Central European Journal of Immunology. 2015 10;40. Available from: http://dx.doi.org/10.5114/ceji.2015.54594.

[61] Cheng Z, l Dai L, Liu Q, Liu M, Wang Q, f Li P, et al. Correlation between polymorphisms in the glucocorticoid receptor gene NR3C1 and susceptibility to asthma in a Chinese population from the Henan Province. Genetics and Molecular Research. 2016;15. Available from: http://dx.doi.org/10.4238/gmr.15028507.

[62] Fu G, Fu L, Cai Y, Zhao H, Fu W. Association between polymorphisms of glucocorticoid receptor genes and asthma: A meta-analysis. Cellular and Molecular Biology. 2018 4;64. Available from: http://dx.doi.org/10.14715/cmb/2018.64.5.3.

[63] Khaleel AS, Falih ES, Hassoon HJ. BclI Polymorphisms of NR3C1 Gene and sST2 Levels Among Asthmatic Patients Under Steroid Treatment. Diyala Journal of Medicine. 2025 12;29. Available from: http://dx.doi.org/10.26505/djm.v29i2.1406.

[64] Kytikova OY, Perelman JM, Novgorodtseva TP, Denisenko YK, Kolosov VP, Antonyuk MV, et al. Peroxisome Proliferator-Activated Receptors as a Therapeutic Target in Asthma. PPAR Research. 2020 1;2020. Available from: http://dx.doi.org/10.1155/2020/8906968.

[65] Banno A, Reddy AT, Lakshmi SP, Reddy RC. PPARs: Key Regulators of Airway Inflammation and Potential Therapeutic Targets in Asthma. Nuclear Receptor Research. 2018;5. Available from: http://dx.doi.org/10.11131/2018/101306.

[66] García-Ricobaraza M, García-Bermúdez M, Torres-Espinola FJ, Moreno MTS, Bleyere MN, Díaz-Prieto LE, et al. Association study of rs1801282 PPARG gene polymorphism and immune cells and cytokine levels in a Spanish pregnant women cohort and their offspring. Journal of Biomedical Science. 2020 11;27. Available from: http://dx.doi.org/10.1186/s12929-020-00694-3.

[67] Koćwin M, Jonakowski M, Majos A, Szemraj J, Kuna P, Panek M. Evaluation of TGF-β isoforms based on selected clinical parameters in asthmatic patients. Alergologia Polska - Polish Journal of Allergology. 2025;12. Available from: http://dx.doi.org/10.5114/pja.2024.138537.

[68] Panek M, Stawiski K, Kaszkowiak M, Kuna P. Cytokine TGFβ Gene Polymorphism in Asthma: TGF-Related SNP Analysis Enhances the Prediction of Disease Diagnosis (A Case-Control Study With Multivariable Data-Mining Model Development). Frontiers in Immunology. 2022 6;13. Available from: http://dx.doi.org/10.3389/fimmu.2022.746360.

[69] Panek MG, Karbownik MS, Górski KM, Koćwin M, Kardas G, Marynowski M, et al. New insights into the regulation of TGF-β/Smad and MPK signaling pathway gene expressions by nasal allergen and methacholine challenge test in asthma. Clinical and Translational Allergy. 2022 7;12. Available from: http://dx.doi.org/10.1002/clt2.12172.

[70] Holster A, Nuolivirta K, Törmänen S, Lauhkonen E, Teräsjärvi J, Vuononvirta J, et al. ¡i¿Interleukin-10¡/i¿ gene polymorphism rs1800896 is associated with post-bronchiolitis asthma at 11–13 years of age. Acta Paediatrica. 2019 5;108. Available from: http://dx.doi.org/10.1111/apa.14838.

[71] Zhou Z, Zhang H, Yuan Y. Association between ¡i¿IL10¡/i¿ rs1800896 polymorphism and risk of pediatric asthma: A meta-analysis. The Clinical Respiratory Journal. 2023 11;17. Available from: http://dx.doi.org/10.1111/crj.13714.

[72] Lauhkonen E, Koponen P, Teräsjärvi J, Gröndahl-Yli-Hannuksela K, Vuononvirta J, Nuolivirta K, et al. IL-10 Gene Polymorphisms Are Associated with Post-Bronchiolitis Lung Function Abnormalities at Six Years of Age. PLOS ONE. 2015 10;10. Available from: http://dx.doi.org/10.1371/journal.pone.0140799.

[73] Mocellin M, de Azeredo Leitão LA, de Araújo PD, Jones MH, Stein RT, Pitrez PM, et al. Association between interleukin-10 polymorphisms and CD4+CD25+FOXP3+ T cells in asthmatic children. Jornal de Pediatria. 2021 9;97. Available from: http://dx.doi.org/10.1016/j.jped.2020.11.008.

[74] Pavón-Romero GF, Pérez-Rubio G, Ramírez-Jiménez F, Ambrocio-Ortiz E, Merino-Camacho CR, Falfán-Valencia R, et al. IL10 rs1800872 Is Associated with Non-Steroidal Anti-Inflammatory Drugs Exacerbated Respiratory Disease in Mexican-Mestizo Patients. Biomolecules. 2020 1;10. Available from: http://dx.doi.org/10.3390/biom10010104.

[75] Zhu Y, Yan X, Zhai C, Yang L, Li M. Association between risk of asthma and gene polymorphisms in CHI3L1 and CHIA: a systematic meta-analysis. BMC Pulmonary Medicine. 2017 12;17. Available from: http://dx.doi.org/10.1186/s12890-017-0515-2.

[76] Chen GL, Wang SC, Shen TC, Chang WS, Lin C, Hsia TC, et al. Significant Association of Chitinase 3-like 1 Genotypes to Asthma Risk in Taiwan. In Vivo. 2021;35. Available from: http://dx.doi.org/10.21873/invivo.12320.

[77] Abe K, Nakamura Y, Yamauchi K, Maemondo M. Role of genetic variations of chitinase 3-like 1 in bronchial asthmatic patients. Clinical and Molecular Allergy. 2018 4;16. Available from: http://dx.doi.org/10.1186/s12948-018-0086-7.

[78] Chen G, Zhang MM, Wang Y, Wu SQ, Wang MG, He JQ. Functional study of the association of ¡i¿CHI3L1¡/i¿ polymorphisms with asthma susceptibility in the Southwest Chinese Han population. Bioscience Reports. 2019 5;39. Available from: http://dx.doi.org/10.1042/bsr20182008.

[79] Yu T, Niu W, Niu H, Duan R, Dong F, Yang T. Chitinase 3-like 1 polymorphisms and risk of chronic obstructive pulmonary disease and asthma in a Chinese population. The Journal of Gene Medicine. 2020 5;22. Available from: http://dx.doi.org/10.1002/jgm.3208.

[80] Khan QU, Bano A, Mazhar I, Asif AB, Tahir MI, Ahmad A, et al. Association of rs7216389 Polymorphism in Gasdermin B (GSDMB) With Childhood Asthma: A Case-Control Study. Cureus. 2025 1. Available from: http://dx.doi.org/10.7759/cureus.76937.

[81] Farzan N, Vijverberg SJ, Hernandez-Pacheco N, Bel EHD, Berce V, Bønnelykke K, et al. 17q21 variant increases the risk of exacerbations in asthmatic children despite inhaled corticosteroids use. Allergy. 2018 7;73. Available from: http://dx.doi.org/10.1111/all.13499.

[82] Hernandez-Pacheco N, Vijverberg SJ, Herrera-Luis E, Li J, Sio YY, Granell R, et al. Genome-wide association study of asthma exacerbations despite inhaled corticosteroid use. European Respiratory Journal. 2020 12;57. Available from: http://dx.doi.org/10.1183/13993003.03388-2020.

[83] Eliasen AU, Pedersen CET, Rasmussen MA, Wang N, Soverini M, Fritz A, et al. Genome-wide study of early and severe childhood asthma identifies interaction between CDHR3 and GSDMB. Journal of Allergy and Clinical Immunology. 2022 9;150. Available from: http://dx.doi.org/10.1016/j.jaci.2022.03.019.

[84] Almacioglu M. Association of childhood asthma with Gasdermin B (GSDMB) and Oromucoid-like 3 (ORMDL3) genes. Northern Clinics of Istanbul. 2023. Available from: http://dx.doi.org/10.14744/nci.2023.22120.

[85] Neville MDC, Choi J, Lieberman J, Duan QL. Identification of deleterious and regulatory genomic variations in known asthma loci. 2018 8. Available from: http://dx.doi.org/10.1101/389031.

[86] Olejnik AE, Kuźnar-Kamińska B. Association of Obesity and Severe Asthma in Adults. Journal of Clinical Medicine. 2024 6;13. Available from: http://dx.doi.org/10.3390/jcm13123474.

[87] Zhu Z, Liu C, Ober C, Camargo CA. Gene–environment interaction at 17q12–q21 locus and its role in asthma pathogenesis. European Respiratory Journal. 2025 10;66. Available from: http://dx.doi.org/10.1183/13993003.01232-2025.

[88] Voorhies K, Mohammed A, Chinthala L, Kong SW, Lee IH, Kho AT, et al. GSDMB/ORMDL3 Rare/Common Variants Are Associated with Inhaled Corticosteroid Response among Children with Asthma. Genes. 2024 3;15. Available from: http://dx.doi.org/10.3390/genes15040420.

[89] Slob EMA, Vijverberg SJH, Ruffles T, Noij LCE, Biermann J, Brouwer AFJ, et al. Genotype-Guided Asthma Treatment Reduces Exacerbations in Children: Meta-Analysis of Two Randomized Control Trials. Allergy. 2024 12;80. Available from: http://dx.doi.org/10.1111/all.16438.

[90] Daya M, Rafaels N, Chavan S, Johnston HR, Shetty A, Gignoux CR, et al. Genome-wide association study of asthma in individuals of African ancestry reveals novel asthma susceptibility loci. 2017 3. Available from: http://dx.doi.org/10.1101/112953.

[91] Washington C, Dapas M, Biddanda A, Magnaye KM, Aneas I, Helling BA, et al. African-specific alleles modify risk for asthma at the 17q12-q21 locus in African Americans. Genome Medicine. 2022 9;14. Available from: http://dx.doi.org/10.1186/s13073-022-01114-x.

[92] Hernandez-Pacheco N, Melén E. Unraveling the genetic architecture of asthma. Annals of Translational Medicine. 2022 12;10. Available from: http://dx.doi.org/10.21037/atm-22-5375.

[93] Shao J, Yang X, Ren D, Luo Y, Lai W. A genetic variation in CHI3L1 is associated with bronchial asthma. Archives of Physiology and Biochemistry. 2019 7;127. Available from: http://dx.doi.org/10.1080/13813455.2019.1634737.

[94] Hansen JW, Thomsen SF, Porsbjerg C, Rasmussen LM, Harmsen L, Johansen JS, et al. YKL-40 and genetic status of¡i¿CHI3L1¡/i¿in a large group of asthmatics. European Clinical Respiratory Journal. 2015 1;2. Available from: http://dx.doi.org/10.3402/ecrj.v2.25117.

[95] Rönnebjerg L, Axelsson M, Kankaanranta H, Backman H, Rådinger M, Lundbäck B, et al. Severe Asthma in a General Population Study: Prevalence and Clinical Characteristics. Journal of Asthma and Allergy. 2021 9;Volume 14. Available from: http://dx.doi.org/10.2147/jaa.s327659.

[96] Kanazawa J, Kitazawa H, Masuko H, Yatagai Y, Sakamoto T, Kaneko Y, et al. A cis-eQTL allele regulating reduced expression of CHI3L1 is associated with late-onset adult asthma in Japanese cohorts. BMC Medical Genetics. 2019 4;20. Available from: http://dx.doi.org/10.1186/s12881-019-0786-y.

[97] Ramphul M, Lo DK, Gaillard EA. Precision Medicine for Paediatric Severe Asthma: Current Status and Future Direction. Journal of Asthma and Allergy. 2021 5;Volume 14. Available from: http://dx.doi.org/10.2147/jaa.s265657.

[98] Masood HA, Othman AM, Al-Sonboli NNA, Ghlab FA, Al-Haidary NM, Al-Shaibani MM. Association of STAT6 gene polymorphism with atopic asthma among Yemeni children in Sana'a city, Yemen. BMC Pediatrics. 2025 5;25. Available from: http://dx.doi.org/10.1186/s12887-025-05710-9.

[99] Ramachandra N, Davoodi P, Mahesh P, Holla A. A preliminary study on the association of single nucleotide polymorphisms of interleukin 4 (IL4), IL13, IL4 receptor alpha (IL4Rα) & Toll-like receptor 4 (TLR4) genes with asthma in Indian adults. Indian Journal of Medical Research. 2015;142. Available from: http://dx.doi.org/10.4103/0971-5916.174551.

[100] Herrera-Luis E, Ortega VE, Ampleford EJ, Sio YY, Granell R, Roos ED, et al. Multi-ancestry genome-wide association study of asthma exacerbations. Pediatric Allergy and Immunology. 2022 6;33. Available from: http://dx.doi.org/10.1111/pai.13802.

[101] Ierodiakonou D, Portelli MA, Postma DS, Koppelman GH, Gerritsen J, t Ten Hacken NH, et al. Urokinase plasminogen activator receptor polymorphisms and airway remodelling in asthma. European Respiratory Journal. 2016 2;47. Available from: http://dx.doi.org/10.1183/13993003.01571-2015.

[102] Kachkovska VV, Kovchun AV, Moyseyenko IO, Dudchenko IO, Prystupa LN. ARG16GLY POLYMORPHISM IN THE B2-ADRENOCEPTOR GENE IN PATIENTS WITH BRONCHIAL ASTHMA. Wiadomości Lekarskie. 2021;74. Available from: http://dx.doi.org/10.36740/wlek202105128.

[103] Mohamed-Hussein AAR, Sayed SS, Eldien HMS, Assar AM, Yehia FE. Beta 2 Adrenergic Receptor Genetic Polymorphisms in Bronchial Asthma: Relationship to Disease Risk, Severity, and Treatment Response. Lung. 2018 9;196. Available from: http://dx.doi.org/10.1007/s00408-018-0153-3.

[104] Keskin O, Ünal Uluca, Birben E, Coşkun Y, Ozkars MY, Keskin M, et al. Genetic associations of the response to inhaled corticosteroids in children during an asthma exacerbation. Pediatric Allergy and Immunology. 2016 5;27. Available from: http://dx.doi.org/10.1111/pai.12566.

[105] Das S, Miller M, Beppu AK, Mueller J, Mcgeough MD, Vuong C, et al. GSDMB induces an asthma phenotype characterized by increased airway responsiveness and remodeling without lung inflammation. Proceedings of the National Academy of Sciences. 2016 10;113. Available from: http://dx.doi.org/10.1073/pnas.1610433113.

[106] Ruan Z, Shi Z, Zhang G, Kou J, Ding H. Asthma susceptible genes in children. Medicine. 2020 11;99. Available from: http://dx.doi.org/10.1097/md.0000000000023051.

[107] Buchheit KM, Sohail A, Hacker J, Maurer R, Gakpo D, Bensko JC, et al. Rapid and sustained effect of dupilumab on clinical and mechanistic outcomes in aspirin-exacerbated respiratory disease. Journal of Allergy and Clinical Immunology. 2022 8;150. Available from: http://dx.doi.org/10.1016/j.jaci.2022.04.007.

[108] Dapas M, Wentworth-Sheilds W, Thompson EE, Kumar R, Lippner E, Wood RA, et al. Cumulative Genetic Risk for Asthma Contributes to Disease Severity in Children with Asthma. 2025 9. Available from: http://dx.doi.org/10.1101/2025.09.08.25335346.

[109] Portelli MA, Rakkar K, Hu S, Guo Y, Adcock IM. Translational Analysis of Moderate to Severe Asthma GWAS Signals Into Candidate Causal Genes and Their Functional, Tissue-Dependent and Disease-Related Associations. Frontiers in Allergy. 2021 10;2. Available from: http://dx.doi.org/10.3389/falgy.2021.738741.

[110] Cha J, Choi S. Gene–Smoking Interaction Analysis for the Identification of Novel Asthma-Associated Genetic Factors. International Journal of Molecular Sciences. 2023 7;24. Available from: http://dx.doi.org/10.3390/ijms241512266.

[111] Shukur W, Alyaqubi K, Dosh R, Al-Ameri A, Al-Aubaidy H, Al-Maliki R, et al. Association of Toll-like receptors 4 (TLR-4) gene expression and polymorphisms in patients with severe asthma. Journal of Medicine and Life. 2021 8;14. Available from: http://dx.doi.org/10.25122/jml-2021-0173.

[112] Martin-Gonzalez E, Perez-Garcia J, Herrera-Luis E, Martin-Almeida M, Kebede-Merid S, Hernandez-Pacheco N, et al. Epigenome-Wide Association Study of Asthma Exacerbations in Europeans. Allergy. 2025 2;80. Available from: http://dx.doi.org/10.1111/all.16490.

[113] Kim DJ, Lim JE, Jung HU, Chung JY, Baek EJ, Jung H, et al. Identification of asthma-related genes using asthmatic blood eQTLs of Korean patients. BMC Medical Genomics. 2023 10;16. Available from: http://dx.doi.org/10.1186/s12920-023-01677-7.

[114] Tse SM, Krajinovic M, Chauhan BF, Zemek R, Gravel J, Chalut D, et al. Genetic determinants of acute asthma therapy response in children with moderate-to-severe asthma exacerbations. Pediatric Pulmonology. 2019 1;54. Available from: http://dx.doi.org/10.1002/ppul.24247.

[115] Shi F, Zhang Y, Qiu C. Gene polymorphisms in asthma: a narrative review. Annals of Translational Medicine. 2022 6;10. Available from: http://dx.doi.org/10.21037/atm-22-2170.

[116] Dijk FN, Vijverberg SJ, Hernandez-Pacheco N, Repnik K, Karimi L, Mitratza M, et al. ¡i¿IL1RL1¡/i¿ gene variations are associated with asthma exacerbations in children and adolescents using inhaled corticosteroids. Allergy. 2019 12;75. Available from: http://dx.doi.org/10.1111/all.14125.